

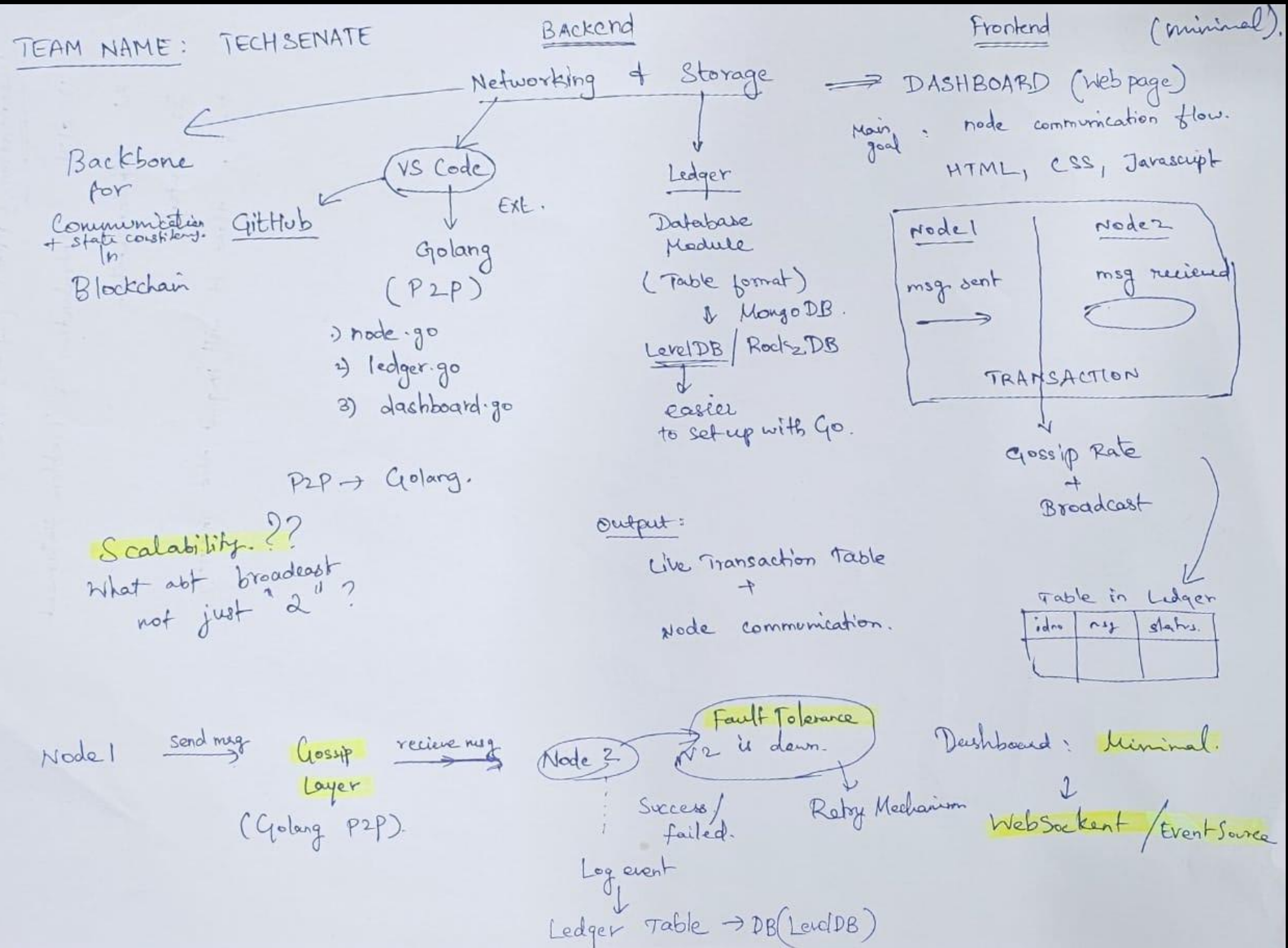
Networking & Storage Prototype using P2P & Ledger Modules

TEAM NAME: TECHSENATE

About Project

- Developed a prototype for the Diamante L1 blockchain stack.
- Networking (P2P): Established a decentralized peer-to-peer network for node communication.
- Storage (Ledger/State Machine): Managed immutable transaction records and maintained accurate state.
- Implemented real-time transaction processing and data persistence.
- Created a visual dashboard to monitor node interactions and ledger updates.

PLAN



Demo Video

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localhost:8080

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P2P Message Visualizer

zuhair

abhijith

Message

Send

Sender Node

Receiver Node

Ledger

Timestamp	Sender	Receiver	Message
2025-05-01T10:41:26+04:00	zuhair	abhijith	hi
2025-05-01T09:21:57+04:00	zuhair	abhijith	great
2025-05-01T00:07:02+04:00	rohit	head	great
2025-04-30T21:58:39+04:00	zuhair	someone	hi
2025-04-30T21:58:27+04:00	zuhair	someone	great
2025-04-30T21:58:11+04:00	zuhair	someone	great
2025-04-30T21:57:54+04:00	zuhair	someone	great
2025-04-30T21:57:41+04:00	zuhair	someone	great
2025-04-30T21:57:11+04:00	zuhair	someone	great

System Components:

- Node 1: Sends transactions/messages.
- Node 2: Receives and stores messages in LevelDB.
- Server: Exposes stored data via REST API.
- Dashboard: Displays live updates of node messages and ledger entries.

Technologies Used:

- GoLang for networking and database operations.
- LevelDB for lightweight key-value storage.
- HTML/CSS/JS for the frontend interface.



Real-world Application

Money Transfer

Recipient Name / ID

Abhi

Amount in USD

400

Send Money

Node 1 (USER)

Receiver Node

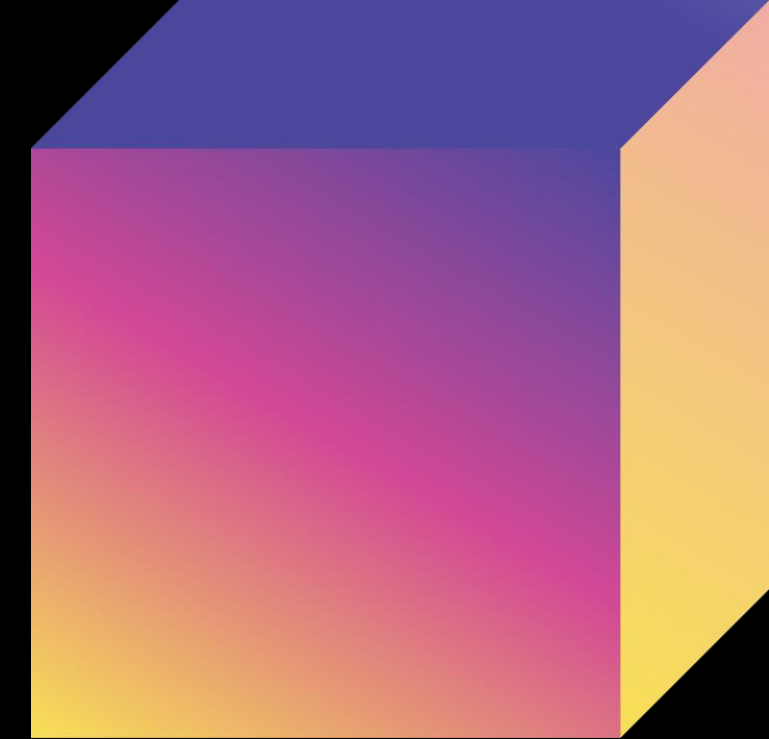
Ledger

From	To	Amount
USER	Abhi	\$200.00
USER	Zuhair	\$200.00
USER	2345674	\$2345.00

Key Features & Outcomes

- Simulated peer-to-peer node communication.
- Real-time gossip message transmission.
- Persistent storage using LevelDB.
- Integrated API and frontend for seamless interaction.
- Modular design separating nodes, server, and dashboard.
- Outcome: Functional minimum viable product demonstrating node communication and ledger operations, providing a foundation for further enhancements like consensus mechanisms and scalability.

Industry Impact



1. Faster & Reliable Blockchain Nodes

What it means: Blockchain networks consist of many nodes (computers) that store and verify transactions.

Impact: Improving their speed and reliability means transactions are processed quicker and the network is more stable.

Why it matters: This is essential for making blockchain systems scalable and user-friendly (e.g., faster payments).

2. Stronger Network Security

DoS Protection: Prevents "Denial of Service" attacks, which try to flood the network with fake traffic and shut it down.

Gossip Protocol: A method nodes use to spread information quickly and efficiently, like how rumors spread in a crowd.

Impact: Together, they make the network more secure and ensure all nodes stay updated reliably.





3. Smarter Ledger Design (using LwDAP)

LwDAP: A framework or method that helps in analyzing and choosing the best structure for the blockchain ledger.

Ledger: A digital record of all transactions in the blockchain.

Impact: Leads to better organization, performance, and adaptability of the blockchain system.

4. Trends & Evolution Insight

What it does: Monitors and analyzes changes in the blockchain industry.

Impact: Helps companies and developers stay updated with new technologies, upgrades, or standards.

Why it matters: Staying ahead ensures competitiveness and innovation.



5. Testing & Simulation

Best-of-breed testing: Using top-quality tools to simulate real-world blockchain behavior.

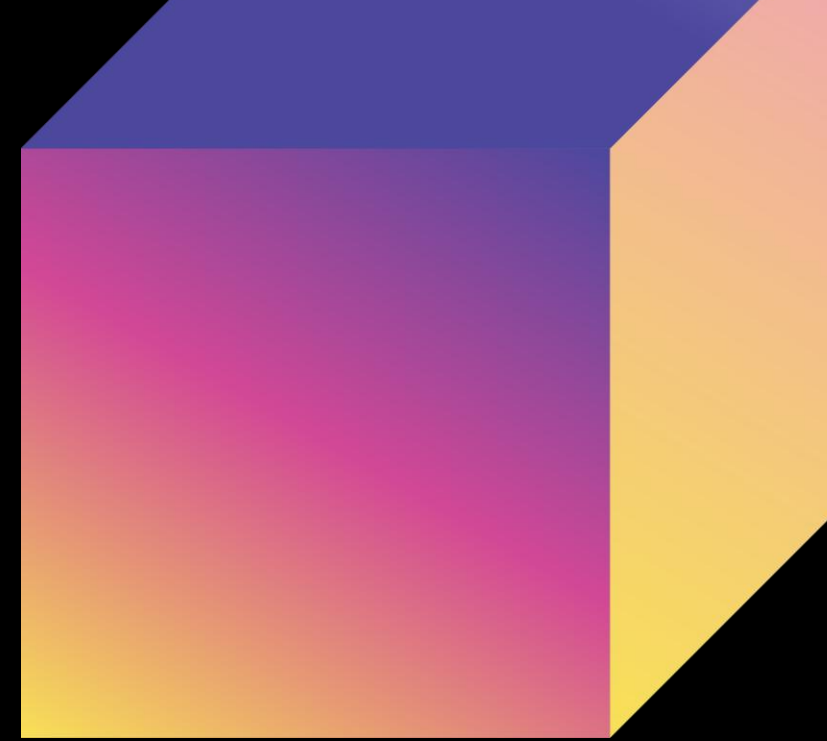
Impact: Lets developers test their networks safely before launch, finding bugs or weaknesses early.

Why it matters: Prevents costly failures and improves user trust.

Web3 is a decentralized version of the internet, where users have control over their data, identities, and assets using blockchain technology.

1. Decentralization- No central authority (like Facebook or Google); instead, apps run on blockchains or peer-to-peer networks.

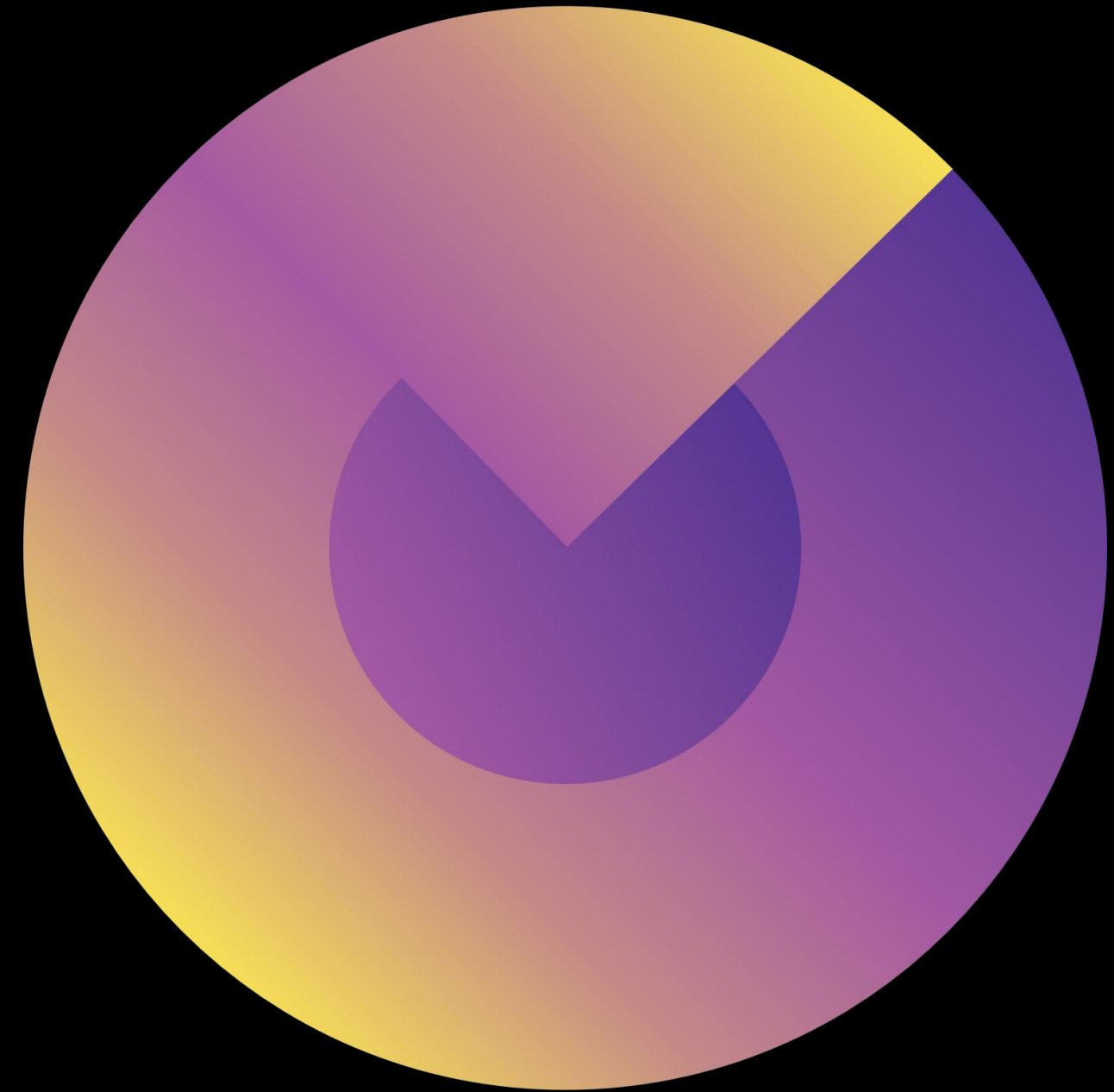
2. Ownership-Users truly own their data, digital items (like NFTs), and money (like crypto wallets).



Thank You

Team Members:

- Zuhair Atham
- Abhijit Lakshmandass
- Anuraag
- Rayn
- Sajjad
- Sahal
- Abdul Kader



Thanks for this amazing opportunity!
Diamante V2 Hackathon